CURRICULUM VITAE – Zikai Zhang

Zikai Zhang

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EDUCATION

Jan. 2023-Present — **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Ph.D. in Computer Science and Engineering

Sep. 2018-Jun. 2021 — **Department of Computer Science and Technology, Huaqiao University, China**
M.Eng. in Computer Technology

Sep. 2014-Jul. 2018 — **Department of Mechatronics and Vehicle Engineering, East China Jiaotong University, China**
B.Eng. in Mechanical Manufacturing and Automation

RESEARCH AND WORK EXPERIENCE

Jan. 2023-Present — **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Graduate Research and Teaching Assistant, Advisor: Dr. Rui (Zoey) Hu
- Large language model fine-tuning in distributed and resource-heterogeneous environments;
- Attack and defense methods for (multimodal) large language models;
- Privacy-preserving and defense mechanisms for resilient distributed machine learning.

Sep. 2022-Nov. 2022 — **Institute of Perceptual Interaction, National Virtual Reality Innovation Center, China**
Algorithm Engineer
- Efficient pupil detection algorithm for virtual-reality glasses.

Aug. 2021-Mar. 2022 — **Institute of Intelligent Vehicle, Tongji University, China**
Research Assistant
- Multimodal visual understanding for autonomous systems.

Apr. 2021-Jul. 2021 — **Department of Research, DMAI, Inc., Guangzhou, China**
Research Intern
- Occlusion and distortion-robust optical character recognition.

Sep. 2019-Jan. 2020 — **Xiamen Institute of Technology, China**
Sessional Lecturer
- Introduce HTML, CSS, JavaScript, and the jQuery framework, and design laboratory sessions.

Apr. 2018-Jun. 2021 — **Computer Vision and Pattern Recognition Lab, Department of Computer Science and Technology, Huaqiao University, China**
Advisor: Dr. Bineng Zhong
M.Eng Thesis: Deep Convolutional Neural Network-based Long-term Object Tracking and Object Detection
- Distractor-aware, efficient, and long-term visual object tracking;
- Semi-supervised learning for object detection.

AWARD AND GRANT

May 2026	**Registration Grant** as a Volunteer, ACL 2026, San Diego, USA
Apr. 2026	**PI,** Google Cloud Research Credits Program, Google, USA.
Mar. 2026	**Early Contributor Badge** for Invited Contributors, Flower Labs, Online.
Oct. 2025	**First Place Poster,** 2025 UNR Cybersecurity Conference, Reno, USA.
Sep. 2025	**Co-PI, Dynamic Language Infrastructure (DLI),** NSF, USA (Under Review).
Aug. 2025	**HDRFS Student Publication & Travel Support,** NSF EPSCoR, USA.
Oct. 2024	**First Place Poster,** 2024 UNR Cybersecurity Conference, Reno, USA.

Jan. 2024	**Co-PI, Nevada WateReuse Research Grant,** WateReuse Association, USA.
Oct. 2023	**2023 IEEE CNS Travel Grant,** NSF, Orlando, USA.
Oct. 2023	**First Place Poster,** 2023 UNR Cybersecurity Conference, Reno, USA.
Oct. 2020	**TOP 1% (5/489)**, **Electric Vehicle Helmet Recognition Competition**, PRCV 2020, Nanjing, China.
Aug. 2020	**TOP 1% (22/2245), Silver Medal**, **Kaggle Global Wheat Detection Competition**, Computer Vision Problems in Plant Phenotyping (CVPPP) Workshop, ECCV 2020, Online.
2018-2021:	**PI, Graduate Innovation Fund** in Scientific Research, Huaqiao University, Xiamen, China.

PUBLICATION AND PATENT

Papers Under Review:

[1] **Zikai Zhang**, Rui Hu, Olivera Kotevska, and Jiahao Xu. *"Vision Token Manipulation Attacks on Cloud-Edge Inference of Large Vision-Language Models."* IEEE Global Communications Conference, GLOBECOM (2026), Under Review.
[2] Jiahao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**. *"CoLa: A Collaborative Learning Framework for Multimodal Large Language Models at the Edge."* IEEE Global Communications Conference, GLOBECOM (2026), Under Review.
[3] **Zikai Zhang**, Rui Hu, Zhiyuan Yu, and Olivera Kotevska. *"STJail: Practical Video Jailbreak Attack Against Multimodal Large Language Models."* The 19th European Conference on Computer Vision, ECCV (2026), Under Review.
[4] **Zikai Zhang**, Rui Hu, Olivera Kotevska, and Jiahao Xu. *"SelfGrader: Stable Jailbreak Detection for Large Language Models using Token-Level Logits."* Conference on Language Modeling, COLM (2026), Under Review.
[5] **Zikai Zhang**, Rui Hu, Ping Liu, and Jiahao Xu. *"Fed-pilot: Optimizing LoRA Allocation for Efficient Federated Fine-Tuning with Heterogeneous Clients."* IEEE Transactions on Neural Networks and Learning Systems, TNNLS, Under Review.
[6] Jiahao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**.*"Majority Bit-Aware Watermarking for Large Language Models."* the Fortieth Annual Conference on Neural Information Processing Systems, NeurIPS (2026), Under Review.

Papers in Refereed Journals and Conferences:

[1] Jiahao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**. *"XMark: Reliable Multi-Bit Watermarking for LLM-Generated Texts."* The 64th Annual Meeting of the Association for Computational Linguistics, **ACL (2026)**.
[2] **Zikai Zhang**, Rui Hu, and Jiahao Xu. *"Heterogeneous Federated Fine-Tuning with Parallel One-Rank Adaptation."* The Fourteenth International Conference on Learning Representations, **ICLR (2026)**.
[3] Jiahao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**. *"Traceable Black-Box Watermarks for Federated Learning."* The Fourteenth International Conference on Learning Representations, **ICLR (2026)**.
[4] **Zikai Zhang**, Suman Rath, Jiahao Xu, and Tingsong Xiao. *"Federated Learning for Smart Grid: A Survey on Applications and Potential Vulnerabilities."* ACM Transactions on Cyber-Physical Systems, **ACM TCPS (2026)**.
[5] Yan Gao, Massimo Roberto Scamarcia, Javier Fernandez-Marques, Mohammad Naseri, Chong Shen Ng, Dimitris Stripelis, Zexi Li, Tao Shen, Jiamu Bai, Daoyuan Chen, **Zikai Zhang**, Rui Hu, InSeo Song, Lee KangYoon, Hong Jia, Ting Dang, Junyan Wang, Zheyuan Liu, Daniel Janes Beutel, Lingjuan Lyu, and Nicholas D Lane. *"FlowerTune: A Cross-Domain Benchmark for Federated Fine-Tuning of Large Language Models."* Advances in Neural Information Processing Systems, **NeurIPS (2025)**.
[6] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"On the Out-of-Distribution Backdoor Attack for Federated Learning."* The 26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing, **ACM MobiHoc (2025)**.
[7] **Zikai Zhang**, Ping Liu, Jiahao Xu, and Rui Hu. *"Fed-HeLLo: Efficient Federated Foundation Model Fine-Tuning with Heterogeneous LoRA Allocation."* IEEE Transactions on Neural Networks and Learning Systems, **TNNLS (2025)**.
[8] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Detecting Backdoor Attacks in Federated Learning via Direction Alignment Inspection."* Proceedings of the Computer Vision and Pattern Recognition Conference, **CVPR (2025)**.

[9] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Achieving Byzantine-Resilient Federated Learning via Layer-Adaptive Sparsified Model Aggregation."* IEEE/CVF Winter Conference on Applications of Computer Vision, **WACV (2025)**.
[10] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Identify Backdoored Model in Federated Learning via Individual Unlearning."* IEEE/CVF Winter Conference on Applications of Computer Vision, **WACV (2025)**.
[11] **Zikai Zhang** and Rui Hu. *"Byzantine-robust Federated Learning with Variance Reduction and Differential Privacy."* 2023 IEEE Conference on Communications and Network Security, **IEEE CNS (2023)**.
[12] Zichen Liang, Hu Cao, Chu Yang, **Zikai Zhang**, and Guang Chen. *"Global-local Feature Aggregation for Event-based Object Detection on Eventkitti."* 2022 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, **IEEE MFI (2022)**.
[13] Siyuan Cheng, Binfei Chu, Bineng Zhong, **Zikai Zhang**, Xin Liu, Zhenjun Tang, and Xianxian Li. *"DRNet: Towards Fast, Accurate and Practical Dish Recognition."* **Science China-Technological Sciences (2021)**.
[14] **Zikai Zhang**, Bineng Zhong, Shengping Zhang, Zhenjun Tang, Xin Liu, and Zhaoxiang Zhang. *"Distractor-Aware Fast Tracking via Dynamic Convolutions and MOT Philosophy."* Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, **CVPR (2021)**.

Patents:

[1] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Qihua Liang, and Xianxian Li. *"Wheat Head Detection Method based on Computer Vision Semi-supervised Pseudo Label Learning."* **CN PATENT** (CN113554627A, 2022), Granted.
[2] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Zhenjun Tang, Xianxian Li, Xin Liu. *"Low-power Consumption Real-time Helmet Detection Method based on Computer Vision Target Detection."* **CN PATENT** (CN113128476A, 2021), Published.

MENTORING AND TEACHING EXPERIENCE

Mentoring:

2025 Fall — **Mentor,** Research Project, **University of Nevada, Reno, USA**
- Students: Mia Fisher, Karam Alkherej
- Project: Implicit Prompt Injection Attack and Defense

2023-2024 — **Mentor,** Capstone Project for Undergraduates, **University of Nevada, Reno, USA**
- Students: Cody Long, Zachary Strazi, Kristian Konstantinov, Jacob Ayers
- Project: Automatic Speech Recognition Attack with Adversarial Examples [Project Page]

2023 Summer — **Mentor,** Research Experiences for Undergraduates (REU), **University of Nevada, Reno, USA**
- Students: Manny Ortiz (Penn State University), Zhang Lin (Colorado School of Mines)
- Project: Backdoor Attacks in Federated Learning Systems (Supported by NSF REU #2150394)

Teaching:

2025 Fall — **Graduate Teaching Assistant,** CS442/642 Cloud Computing, **University of Nevada, Reno, USA**
- Design course projects and tutorials on AWS-powered AI web applications;
- Lead office hours and discussion sessions to support students with programming assignments and course concepts.

2025 Spring — **Graduate Teaching Assistant,** CS302 Data Structures, **University of Nevada, Reno, USA**
- Lead office hours to support students with programming assignments and conceptual questions;
- Grade programming assignments and quizzes, and provide constructive feedback on algorithm design and code quality.

2019-2020 — **Sessional Lecturer,** Web Interaction Design, **Xiamen Institute of Technology, China**
- Introduce HTML, CSS, JavaScript, and the jQuery framework;
- Design and conduct laboratory sessions;
- Develop and grade quizzes, midterm exams, and final exams.

PROFESSIONAL SERVICE

Journal Reviewer:

IEEE Transactions on Neural Networks and Learning Systems

IEEE Transactions on Dependable and Secure Computing

IEEE/ACM Transactions on Networking

Transactions on Machine Learning Research

Information Processing and Management

IEEE Open Journal of the Communications Society

IEEE Industrial Electronics Magazine

IEEE Latin America Transactions

Pattern Recognition

China Communication

Scientific Reports

npj Artificial Intelligence

Cluster Computing

Journal of King Saud University Computer and Information Sciences

Journal of Information Security and Applications

The Journal of Supercomputing

Journal of Real-Time Image Processing

Engineering, Technology & Applied Science Research

International Journal of Machine Learning and Cybernetics

TPC/PC Member:

The Annual AAAI Conference on Artificial Intelligence (AAAI) 2026

International Conference on Smart Mechatronics (ICSMech) 2024

IEEE Consumer Communications & Networking Conference (CCNC) 2024

Conference Reviewer:

The International Conference on Learning Representations (ICLR) 2026

International Conference on Machine Learning (ICML) 2025, 2026

Conference on Neural Information Processing Systems (NeurIPS) 2022, 2024, 2025

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025, 2026

International Conference on Computer Vision (ICCV) 2025

The European Conference on Computer Vision (ECCV) 2026

Conference on Language Modeling (COLM) 2026

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025

International Conference on Pattern Recognition (ICPR) 2025

IEEE Consumer Communications & Networking Conference (CCNC) 2025

IEEE International Conference on Computer Communications and Networks (ICCCN) 2025

CVPR Workshop on Federated Learning for Computer Vision (FedVision) 2026